

# ASSIGNMENT-6.5

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Task Description #1 (AI-Based Code Completion for Conditional Eligibility Check)

Task: Use an AI tool to generate eligibility logic.

Prompt:

“Generate Python code to check voting eligibility based on age and citizenship.”

Expected Output:

- AI-generated conditional logic.
- Correct eligibility decisions.
- Explanation of conditions.

**CODE:**

```
def check_voting_eligibility(age: int, citizenship: str) -> bool:
    """
    Returns True if the person is eligible to vote.

    Eligibility: age >= 18 AND citizenship matches required country or 'yes' for
    citizen flag.
    """

    # Normalize inputs
    citizenship = citizenship.strip().lower()

    # Check age first (fast fail)
    if age < 0:
        raise ValueError("Age cannot be negative.")

    if age < 18:
```

```

    return False

# Check citizenship: allow either exact country name or boolean-like answers
allowed_yes = {"yes", "y", "true", "citizen"}

if citizenship in allowed_yes:

    return True

# If user provided a country name, replace "yourcountry" with the real
country you require

required_country = "india" # <-- change as needed

if citizenship == required_country:

    return True

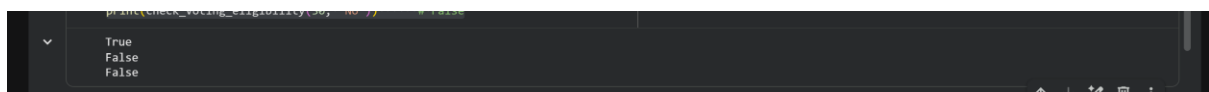
return False

# Example usages / tests

print(check_voting_eligibility(20, "India")) # True (if required_country is india)
print(check_voting_eligibility(17, "yes"))   # False
print(check_voting_eligibility(30, "No"))    # False

```

## OUTPUT:



```

True
False
False

```

## EXPLANATION:

- We normalize citizenship to make checks robust.
- We validate age  $\geq 0$ . AI-generated snippets often forget input normalization and negative-age checks.
- Replace `required_country` with your country or provide an explicit policy (the assignment likely expects a simple age  $\geq 18$  and citizenship == "Yes" approach — above is a slightly more robust version).

## Task Description #2(AI-Based Code Completion for Loop-Based String Processing)

Task: Use an AI tool to process strings using loops.

Prompt:

“Generate Python code to count vowels and consonants in a string using a loop.”

Expected Output:

- AI-generated string processing logic.
- Correct counts.
- Output verification.

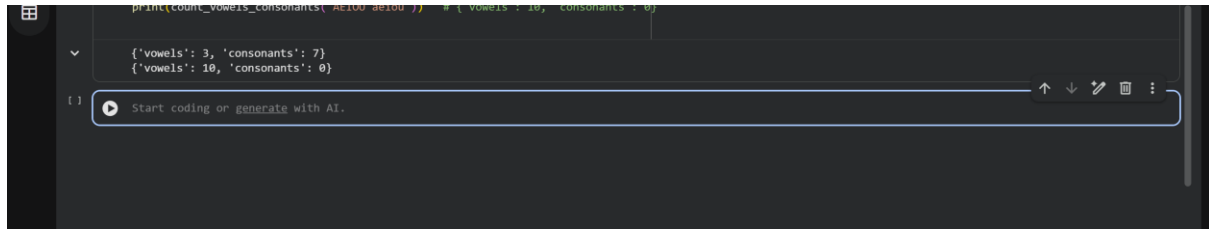
### **CODE:**

```
def count_vowels_consonants(s: str) -> dict:
    s = s.lower()
    vowels = set("aeiou")
    vowel_count = 0
    consonant_count = 0
    for ch in s:
        if ch.isalpha():
            if ch in vowels:
                vowel_count += 1
            else:
                consonant_count += 1
    return {"vowels": vowel_count, "consonants": consonant_count}

# Example tests
print(count_vowels_consonants("Hello World!")) # {'vowels': 3, 'consonants': 7}
```

```
print(count_vowels_consonants("AEIOU aeiou")) # {'vowels': 10, 'consonants': 0}
```

### OUTPUT:



```
print(count_vowels_consonants("AEIOU aeiou")) # {'vowels': 10, 'consonants': 0}
```

```
{  
  'vowels': 10, 'consonants': 0  
}
```

### EXPLANATION:

- Use `str.isalpha()` to ignore digits/punctuation — many AI outputs forget this and count punctuation as consonants or letters.
- The loop-based method explicitly shows the algorithm (as required by the assignment). If you want to be concise, a comprehension or `sum(...)` could be used, but the loop is clearer for demonstration.

### Task Description #3 (AI-Assisted Code Completion Reflection

Task)

Task: Use an AI tool to generate a complete program using classes, loops, and conditionals.

Prompt:

“Generate a Python program for a library management system using classes, loops, and conditional statements.”

Expected Output:

- Complete AI-generated program.
- Review of AI suggestions quality.
- Short reflection on AI-assisted coding experience

### CODE:

class Book:

```
def __init__(self, book_id: int, title: str, author: str):  
    self.book_id = book_id
```

```

        self.title = title

        self.author = author

        self.is_borrowed = False

    def __str__(self):

        status = "Borrowed" if self.is_borrowed else "Available"

        return f"[{self.book_id}] {self.title} by {self.author} - {status}"

class Library:

    def __init__(self):

        self.books = {} # book_id -> Book

        self.next_id = 1

    def add_book(self, title: str, author: str):

        b = Book(self.next_id, title, author)

        self.books[self.next_id] = b

        self.next_id += 1

        return b.book_id

    def list_books(self):

        return list(self.books.values())

    def borrow_book(self, book_id: int) -> bool:

        book = self.books.get(book_id)

        if not book:

            return False # not found

        if book.is_borrowed:

            return False # already borrowed

        book.is_borrowed = True

        return True

    def return_book(self, book_id: int) -> bool:

```

```

    book = self.books.get(book_id)

    if not book or not book.is_borrowed:

        return False

    book.is_borrowed = False

    return True

def run_library_cli():

    lib = Library()

    # add some sample books

    lib.add_book("1984", "George Orwell")

    lib.add_book("To Kill a Mockingbird", "Harper Lee")

    while True:

        print("\nLibrary Menu:")

        print("1. List books")

        print("2. Add book")

        print("3. Borrow book")

        print("4. Return book")

        print("5. Exit")

        choice = input("Choose an option: ").strip()

        if choice == "1":

            for b in lib.list_books():

                print(b)

        elif choice == "2":

            title = input("Title: ").strip()

            author = input("Author: ").strip()

            book_id = lib.add_book(title, author)

            print(f"Added with id {book_id}")

```

```

elif choice == "3":
    try:
        bid = int(input("Book id to borrow: ").strip())
    except ValueError:
        print("Invalid id.")
        continue

    ok = lib.borrow_book(bid)

    print("Borrowed." if ok else "Cannot borrow (not found or already
borrowed).")

elif choice == "4":
    try:
        bid = int(input("Book id to return: ").strip())
    except ValueError:
        print("Invalid id.")
        continue

    ok = lib.return_book(bid)

    print("Returned." if ok else "Cannot return (not found or not
borrowed).")

elif choice == "5":
    print("Goodbye.")
    break
else:
    print("Invalid option. Try again.")

# To run the CLI, uncomment below:

if __name__ == "__main__":
    run_library_cli()

```

**OUTPUT:**

```
Library Menu:
1. List books
2. Add book
3. Borrow book
4. Return book
5. Exit
Choose an option: 1
[1] 1984 by George Orwell - Available
[2] To Kill a Mockingbird by Harper Lee - Available

Library Menu:
1. List books
2. Add book
3. Borrow book
4. Return book
5. Exit
Choose an option: 1
[1] 1984 by George Orwell - Available
[2] To Kill a Mockingbird by Harper Lee - Available

Library Menu:
1. List books
2. Add book
3. Borrow book
4. Return book
5. Exit
Choose an option: 5
Goodbye.
```

## EXPLANATION:

- The solution uses Book and Library classes, demonstrates loops (menu loop) and conditionals for option handling.
- AI tends to produce huge monolithic programs — prefer small, testable methods (add\_book, borrow\_book) which are easy to unit test.
- Test ideas: borrow a non-existent book, borrow twice, return without borrowing.

## Task Description #4 (AI-Assisted Code Completion for Class-Based Attendance System)

Task: Use an AI tool to generate an attendance management class.

Prompt: “Generate a Python class to mark and display student attendance using loops.”

Expected Output:

- AI-generated attendance logic.
- Correct display of attendance.
- Test cases

## CODE:

class Attendance:

```
def __init__(self, students: list):
    # students: list of student names or ids
    self.students = list(students)
```



```
# Initialize attendance map: student -> list of dates (or boolean per session)
```

```
self.records = {s: [] for s in self.students}
```

```
def mark_attendance(self, date: str, present_list: list):
```

```
    """
```

```
    date: string like '2026-01-23'
```

```
    present_list: list of student names/ids who are present
```

```
    """
```

```
    present_set = set(present_list)
```

```
    for s in self.students:
```

```
        self.records[s].append((date, s in present_set))
```

```
def display_attendance(self):
```

```
    # Pretty print summary: student -> attendance count / total sessions
```

```
    total_sessions = 0
```

```
    # infer total sessions from first student (or 0)
```

```
    if self.students:
```

```
        total_sessions = len(self.records[self.students[0]])
```

```
    print(f"Total sessions: {total_sessions}")
```

```
    for s in self.students:
```

```
        present_count = sum(1 for _, present in self.records[s] if present)
```

```
        print(f"{s}: {present_count}/{total_sessions} present")
```

```
def get_attendance_percentage(self, student):
```

```
    sessions = self.records.get(student, [])
```

```
    if not sessions:
```

```
        return 0.0
```

```
    present_count = sum(1 for _, present in sessions if present)
```

```
    return (present_count / len(sessions)) * 100.0
```

# Test cases

```
students = ["Alice", "Bob", "Charlie"]
```

```
att = Attendance(students)
```

```
att.mark_attendance("2026-01-20", ["Alice", "Charlie"])
```

```
att.mark_attendance("2026-01-21", ["Alice"])
```

```
att.mark_attendance("2026-01-22", ["Bob", "Alice"])
```

```
att.display_attendance()
```

```
print("Alice %:", att.get_attendance_percentage("Alice"))
```

```
print("Bob %:", att.get_attendance_percentage("Bob"))
```

## OUTPUT:

A terminal window with a dark background. On the left, there are icons for a file explorer and a list. The output text is as follows:

```
Total sessions: 3  
Alice: 3/3 present  
Bob: 1/3 present  
Charlie: 1/3 present  
Alice %: 100.0  
Bob %: 33.33333333333333
```

## EXPLANATION:

- records stores tuples (date, present\_flag) so you can later expand to reasons, late marks, etc.
- Tests added demonstrate marking multiple sessions and computing percentages.
- AI outputs sometimes use parallel lists incorrectly; prefer dictionary keyed by student for clarity.

## Task Description #5 (AI-Based Code Completion for Conditional

### Menu Navigation)

Task: Use an AI tool to complete a navigation menu.

Prompt: "Generate a Python program using loops and conditionals to simulate an ATM menu."

Expected Output:

- AI-generated menu logic.
- Correct option handling.

- Output verification

**CODE:**

```
def atm_cli():  
    balance = 1000.0 # starting balance  
  
    while True:  
        print("\nATM Menu:")  
        print("1. Check Balance")  
        print("2. Deposit")  
        print("3. Withdraw")  
        print("4. Exit")  
        choice = input("Choose: ").strip()  
        if choice == "1":  
            print(f"Balance: {balance:.2f}")  
        elif choice == "2":  
            try:  
                amt = float(input("Enter deposit amount: ").strip())  
                if amt <= 0:  
                    print("Enter a positive amount.")  
                else:  
                    balance += amt  
                    print("Deposit successful.")  
            except ValueError:  
                print("Invalid amount.")  
        elif choice == "3":  
            try:  
                amt = float(input("Enter withdrawal amount: ").strip())
```

```

    if amt <= 0:
        print("Enter a positive amount.")

    elif amt > balance:
        print("Insufficient funds.")

    else:
        balance -= amt
        print("Withdrawal successful.")

except ValueError:
    print("Invalid amount.")

elif choice == "4":
    print("Thank you. Exiting.")
    break

else:
    print("Invalid option. Try again.")

# To run ATM, uncomment:
# if __name__ == "__main__":
#     atm_cli()

```

## OUTPUT:

```

ATM Menu:
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Choose: 1
Balance: 1000.00

ATM Menu:
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Choose: 3
Enter withdrawal amount: 200
Withdrawal successful.

ATM Menu:
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Choose: 1
Balance: 800.00

ATM Menu:
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Choose: 5
Invalid option. Try again.

ATM Menu:
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Choose: 5
Invalid option. Try again.

ATM Menu:
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Choose: 4
Thank you. Exiting.

```

**EXPLANATION:**

- This is a standard menu loop with input validation.
- Important AI pitfalls: forgetting to validate numeric input or allowing negative deposits/withdrawals.